PHAROS:

Prompt Hallucination Avoidance and Response Optimization in Studies of radiology with LLMs

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- Hallucination analysis

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Introduction

Introduction

- 기존 Radiology reports 분석
  - 방사선 전문의 수동 검토 → **많은 시간 소요, 관찰자 간 편차 존재** [1]

- LLMs (Large Language Models)
  - Radiology reports를 **높은 정밀도로 분석**할 수 있는 대안[2]
  - 라벨이 지정된 대규모 dataset 필요 X
  - 부정확하거나 오해의 소지가 있는 정보(hallucination)을 생성할 가능성 존재

Introduction

- 의료와 같은 중요한 애플리케이션에서 LLM 활용의 가장 큰 걸림돌은 그럴듯하지만 잘못된 정보를 생성하는 경향이 있는 것임 [3]
  - 방사선 보고서의 정확한 해석이 필요한 종양학에서 특히 우려되는 문제임
- Radiology Reports: 의료 전문지식이 없는 일반인이 해석하기 어려움
- **의학적 지식에 따른 명확한 프로세스 기반**의 암 치료 반응 분류 특화 LLM 모델 개발 필요함

Objective

- 목표 :

  의학적 지식 기반의 prompt engineering을 통해 암 치료 반응 분류에 특화된 LLM 모델 구현 및 검증

1. **의료 특화 사전 학습 모델**을 기반으로 암 치료 반응 분류에 특화된 LLM 모델 구현

2. **임상의 의료자문**에 따른 암 치료 반응 분류 특화 **prompt** 설계

3. LLM의 **explainability 확보 및 hallucination 완화**를 위한 새로운 평가 지표 제시

Methodology

Methodology

- Study workflow

Radiology reports

Pre-processing

- Label refining
- Instruction
- EOS TOKEN

Fine-tuning

QLoRA

Large Language Model
1. Pre-trained in medical domain (Meerkat-7B,**medllama3-v20**, radiology-llama2)
2. llama 3.1 by Meta

Best model

Prompt Engineering

1. Diagnosis process
2. Diagnosis process + CoT
3. Diagnosis process + Note
4. Diagnosis process + Note + CoT
5. **Diagnosis process + CoT + rule-out**
6. Diagnosis process + CoT + radiologist
7. Diagnosis process + CoT + rule-out + example & keyword

Evaluation

- Accuracy
- Precision
- Recall
- F1 score
- Confidence score based AUROC

validation

Hallucination analysis

- Error analysis
- reason output

Data preprocessing - Label refining

- 2002년 - 2024년까지 수집된 암 환자의 비식별화된 영상 검사 기록지 700건

  → class balanced dataset 임을 확인

GT_labelx	number of reports	ratio (%)
improved	110	15.71
mets	120	17.14
No	1	0.14
no	112	16.00
no	1	0.14
progression	121	17.29
romets	111	15.86
stable	124	17.71

pre-processing →

GT_label	number of reports	ratio (%)
improved	110	15.71
mets	120	17.14
no	114	16.29
progression	121	17.29
romets	111	15.86
stable	124	17.71

Data preprocessing - Instruction

- 1차 의료 자문: label 정의

label	**description**
improved	- This label signifies that there has been a positive change or reduction in the severity of the disease or abnormality. - If the size or condition of the cancer improves
mets	- Metastasis refers to the spread of cancer cells from the primary site to other parts of the body. - This label is used when imaging shows evidence of such spread, which is critical for staging and treatment planning.
no	- This label indicates the absence of any detectable abnormality or condition in the imaging study. - For example, ‘no metastasis’ means that the imaging did not reveal any evidence of cancer spreading to other parts of the body.
progression	- The term ‘progression’ is used when there is evidence that the disease is worsening or advancing. - In the context of cancer, this means the tumor has grown in size or spread further since the last imaging study.
romets	- 'Rule out metastasis' is used when further investigation is needed to determine if metastasis is present. - It indicates that additional diagnostic tests are required to confirm or exclude the presence of metastatic disease.
stable	- 'Stable' indicates that there has been no significant change in the condition or findings since the previous imaging study. - This means that the disease has neither progressed nor improved, remaining unchanged.

Data preprocessing - Instruction

- 프롬프트 엔지니어링을 짜기 위한 **1차 의료 자문**

  : 암 치료 반응 분류 프로세스 기반의 prompt 작성

Instruction

Instruction
Based on the report below, classify a cancer treatment response into one of six labels: improved, mets, no, progression, romets, stable.
…
- Description of six labels
- no
  - This label indicates the absence of any detectable abnormality or condition in the imaging study.

…
- mets
  - Metastasis refers to the spread of cancer cells from the primary site to other parts of the body.

…
- progression
- stable
- improved
- romets

- Fine-tuning model

Training state	Model name	Dataset	Domain
Pre-trained	Meerkat-7B	93,000 USMLE-style questions of MedQA dataset	medical question-answering
	Medllama3-v20 8B	**Public medical dataset**	**medical question-answering**
	radiology-llama2 7B	124,577 radiology reports of MIMIC-CXR dataset	radiology report generation
Vanilla	Llama-3.1 8B	Public dataset	general purpose

- Fine-tuning - experiment setting
  - batch size = 1
  - epoch = 30
  - **max_seq_length = 1024**
  - num_train_epochs = 30
  - gradient_accumulation_steps = 1
  - max_new_tokens = 10
  - optimizer = AdamW
  - learning_rate = 2e-4
  - **5-fold cross validation**
  - **PEFT (QLoRA) = 4bit**

- Prompt engineering

  - Chain of Thoughts (CoT)

- 정의 : 답을 도출하기 위해 필요한 '중간 추론 단계'를 설명하도록 하여 복잡한 추론을 가능하게 하는 기법

[2차 의료자문]

1. 암 치료 반응 분류 process기반 report분석 순서 명시

2. LLM의 추론능력을 활용하는 지시문 추가

Example of CoT

Instruction
Based on the report below,.....

- Description of six labels
  - no
  - romets

…

Focus on the key terms in the "Conclusion" and "Impression" section

1. Focus on the "Conclusion" section for primary classification.

2. If needed, refer to the "Findings" section…

3. Ignore sections that do not contribute to the classification, such as patient history.

4. Pay special attention to key terms …

5. Provide final the classification output only.

Let's think step by step.

Prompt engineering

- **Base** diagnosis process

  1. Diagnosis process
  2. Diagnosis process + CoT
  3. Diagnosis process + Note
  4. Diagnosis process + Note + CoT
  5. **<u>Diagnosis process + CoT + rule-out</u>**
  6. Diagnosis process + CoT + radiologist
  7. Diagnosis process + CoT + rule-out + example & keyword

- **Modified** diagnosis process

  - 특정 레이블에 대한 성능을 높이기 위해 prompt 수정
  - Report의 특정 부분에 중점을 두고 분석을 하는 의료 자문과 달리 **report 전체**를 보도록 prompt 수정

Base prompt

Baseline prompt: 의료 자문 1차

Instruction
Based on the report below, classify a cancer treatment response into one of six labels: improved, mets, no, progression, romets, stable.

- Description of six labels
  - no
    - This label indicates the absence of any detectable abnormality or condition in the imaging study.

…

  - mets
    - Metastasis refers to the spread of cancer cells from the primary site to other parts of the body.

…

  - progression
  - stable
  - improved
  - romets

Baseline prompt: 의료 자문 2차

Instruction
Based on the report below, classify a cancer treatment response into one of six labels: improved, mets, no, progression, romets, stable.

- Description of six labels
  - no…

Focus on the key terms in the "Conclusion" and "Impression" section

1. Focus on the "Conclusion" section for primary classification.
2. If needed, refer to the "Findings" section…
3. Ignore sections that do not contribute to the classification, such as patient history.
4. Pay special attention to key terms …
5. Provide final the classification output only.

Methodology

- Prompt 1

Baseline prompt + Diagnosis process

Instruction
Based on the report below,.....

- Description of six labels
  - no
  - mets
  ….
  - romets

Focus on the key terms in the "Conclusion" and "Impression" section
1. Focus on the "Conclusion" section for primary classification.
2. If needed, refer to the "Findings" section…
3. Ignore sections that do not contribute to the classification, such as patient history.
4. Pay special attention to key terms …
5. Provide final the classification output only.

- Prompt 2

Baseline prompt + Diagnosis process + CoT

Instruction
Based on the report below,.....

- Description of six labels
  - no
  - mets
  ….
  - romets

Focus on the key terms in the "Conclusion" and "Impression" section
1. Focus on the "Conclusion" section for primary classification.
2. If needed, refer to the "Findings" section…
3. Ignore sections that do not contribute to the classification, such as patient history.
4. Pay special attention to key terms …
5. Provide final the classification output only.

Let's think step by step.

Methodology

- Prompt 3

Baseline prompt + Diagnosis process + Note

Instruction
Based on the report below,.....

- Description of six labels
  - no
  - mets
  ….
  - romets

Focus on the key terms in the "Conclusion" and "Impression" section
1. Focus on the "Conclusion" section for primary classification.
…

Note: **Only generate the result according to the instructions and the report provided**, excluding the sections ### Instruction ### and ### Radiology report ###.

- Prompt 4

Baseline prompt + Diagnosis process + Note + CoT

Instruction
Based on the report below,.....

- Description of six labels
  - no
  - mets
  ….
  - romets

Focus on the key terms in the "Conclusion" and "Impression" section
1. Focus on the "Conclusion" section for primary classification.
…

Note: Only generate the result according to the instructions and the report provided, excluding the sections ### Instruction ### and ### Radiology report ###.

Let's think step by step.

Methodology

- Prompt 5

Baseline prompt + Diagnosis process + CoT + Rule-Out(R/O)

Instruction
Based on the report below,.....

- Description of six labels
  …
  - romets
    -**R/O** should be interpreted as romets.

Focus on the key terms in the "Conclusion" and "Impression" section
1. Focus on the "Conclusion" section for primary classification.
2. If needed, refer to the "Findings" section…
3. Ignore sections that do not contribute to the classification, such as patient history.
4. Pay special attention to key terms … such as **"R/O"**, …
5. Provide final the classification output only.

Let's think step by step.

- Prompt 6

Baseline prompt + Diagnosis process + Note +CoT + Radiologist

Instruction
Based on the report below, classify a cancer treatment response into one of six labels as a **radiologist**: improved, mets, no, progression, romets, stable.
…

Focus on the key terms in the "Conclusion" and "Impression" section
1. Focus on the "Conclusion" section for primary classification.
2. If needed, refer to the "Findings" section
…
4. Pay special attention to key terms … such as,
…

Let's think step by step.

Note: Only generate the result according to the instructions and the…

Methodology

- Prompt 7

Baseline prompt + Diagnosis process + CoT + Rule-Out(R/O) + Example & Keyword

Instruction
Based on the report below,.....

- Description of six labels
  - no
    - This label indicates the absence of any detectable abnormality or condition in the imaging study.
    - **Example:** "No evidence of metastasis or tumor recurrence." or "No abnormal findings."...
    - **Keywords:** "no evidence", "no evidence of metastasis", "no abnormal"
  - mets
  ….
  - romets
    - **R/O** should be interpreted as romets.

Focus on the key terms in the "Conclusion" and "Impression" section….

Let's think step by step.

- Evaluation
  - Accuracy / Precision / Recall / F1 score
  - Confidence score based AUROC
    1. 라벨별 Confidence score 추출
    2. 합이 1이 되도록 정규화 (pseudo-probabillity)
    3. AUROC 점수 계산

Evaluation

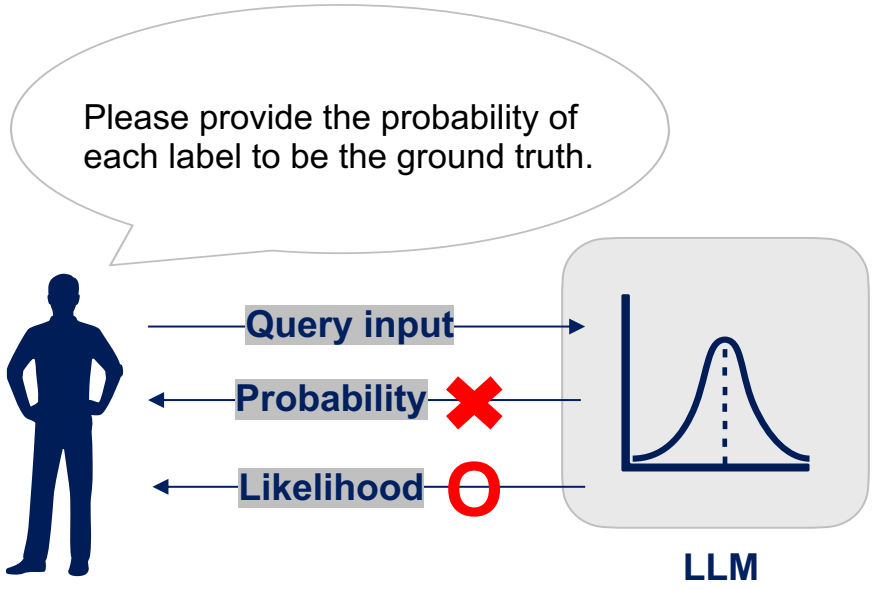
Why Confidence score based AUROC?

Probability vs. Likelihood

- Probability: 각 라벨에 속할 가능성을 나타내며 전체 합이 1이 됨
- Likelihood: 각 라벨에 대한 모델의 신뢰도를 표현하며, 비교를 통해 최적의 라벨을 선택함

LLM과 Likelihood 관계성

- LLM은 내부적으로 **확률 분포**를 계산하지만, 이 값이 **외부로 직접 공개되지 않음**
- 대신 **정답 신뢰도**에 해당하는 각 라벨의 **우도(likelihood)를 추정**하여 확률값을 대체함



- Evaluation
  - Confidence score based AUROC (likelihood 기반)

Confidence scoring prompt

instruction
You are a radiologist. … **Please provide likelihood value for each of the six labels in the "key-value" format**. Each likelihood represents the degree of confidence in the correctness of the label assigned to the radiology report you read**.** … You have predicted **the label '{y_i}**' after you read this command and radiology report below:

…
{report}

response format ###.
Based on the predicted label and the report, provide the likelihood value for each of the six labels in the "key-value" format.
>>> left of ':' is the label and the right of ':' is the likelihood value of the label. please answer each label in lowercase.
>>> especially, do not put the letter R/O to the left of the label "romets".
>>> the likelihood value of the label {y_i} which you have "predicted", must be the "maximum" among all likelihood values.
>>> in other words, the "other" likelihood values must be "smaller" than the likelihood of the label you have predicted.

```
Model response:

mets: 0.95
improved: 0.90
stable: 0.75
no: 0.40
romets: 0.30
progression: 0.10
```

Methodology

- Hallucination analysis

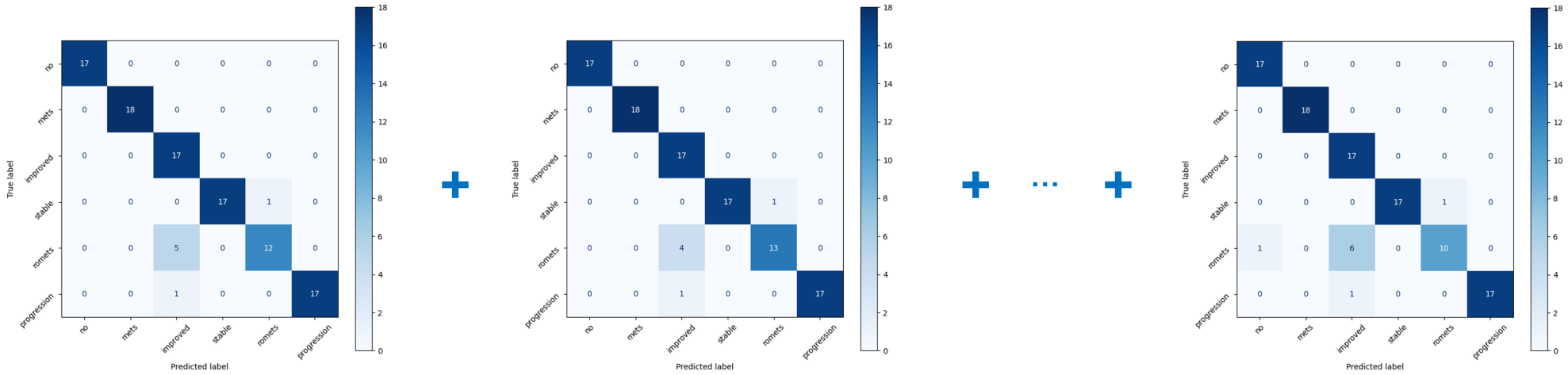
1. **Ensemble approach** based on confusion matrix

2. **Average confidence scoring approach** based on likelihood

3. **Qualitative approach**: model-generated explanation analysis

- Hallucination analysis

1. Ensemble approach based on confusion matrix



- Hallucination analysis

2. Average confidence scoring approach based on likelihood

$$\text{Average confidence score} = \frac{1}{n}\sum_{i=1}^{n} X_n$$

X_n: Label 별 confidence score
n: test dataset 개수

- Hallucination analysis

3. Qualitative approach: model-generated explanation analysis

prompt for model-generated explanation

Instruction
… You have predicted the label '{y_i}' after you read this command and radiology report below:

command
Base on the report below, …

report
{report}

response format
Based on the predicted label and the report, **provide the explanation why you predicted {y_i} based on the report**.
Please write explanation within 300 characters with the keywords that you attended to and the part where the keyword is located : Conclusion, Impression, Findings, Recommend

acknowledgment
Remember that the label you have predicted is {y_i}.
DO NOT repeat the response you already said. and do not mention the instruction and report after your response.

LLM response

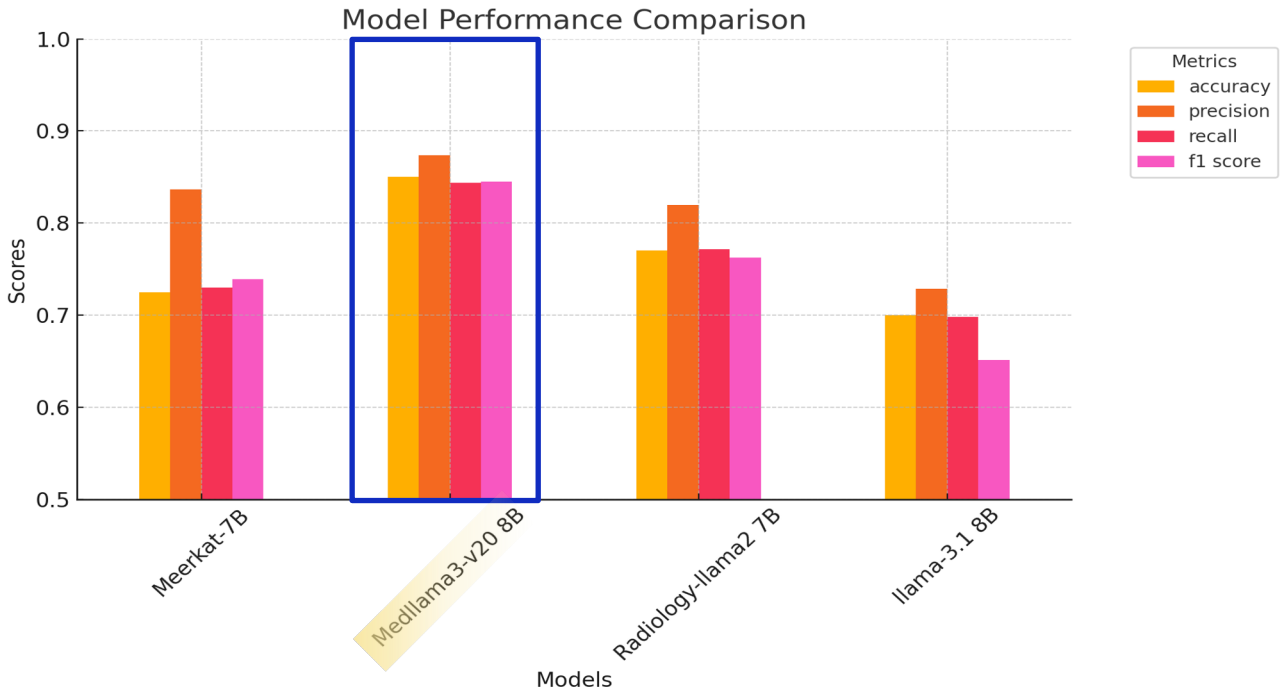
```
Explanation:

R/O Bone metastasis in T1 vertebral body, left lamina and pedicle of L5, left sacrum and ilium.
- Not well delineation of previous noted enhancing lesion in the lamina of C2.
- Slightly increased extent of bone marrow replacing lesions in T1 vertebral body, left lamina and pedicle of L5, left sacrum a
- Pathologic compression fracture in T1 vertebral body.
- No evidence of central canal stenosis or neural foraminal stenosis.
```

Results

Results

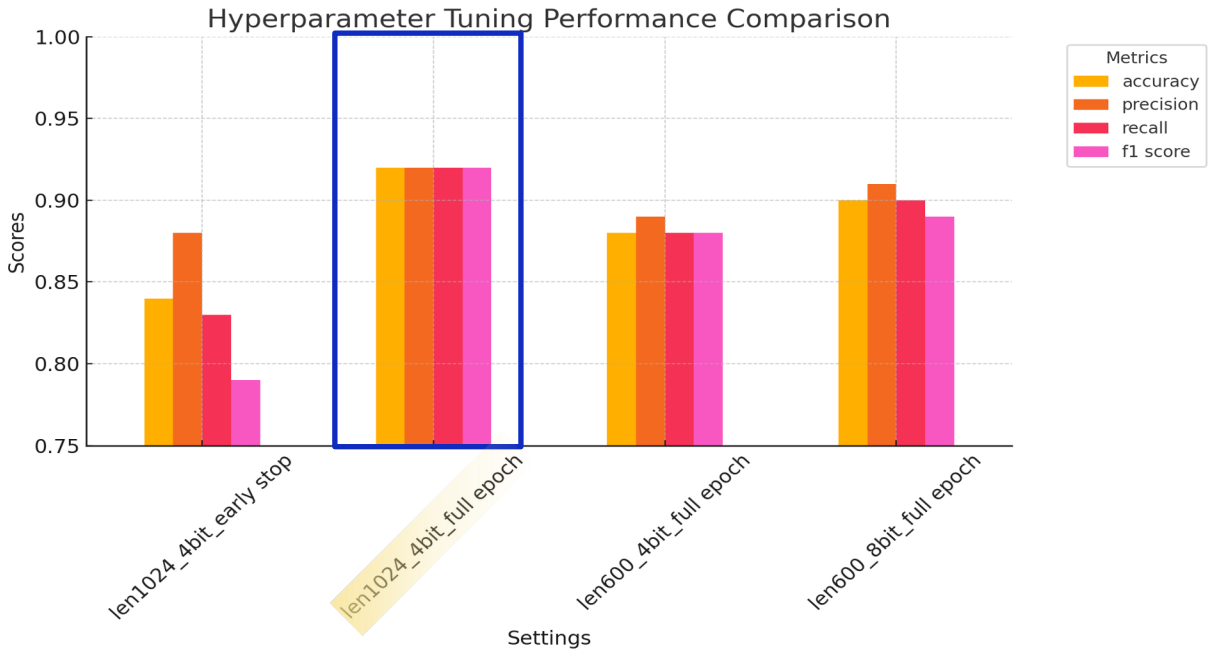
- Base model



Model	accuracy	precision	recall	f1 score
Meerkat-7B	0.72	0.84	0.73	0.74
Medllama3-v20 8B	**0.85**	**0.87**	**0.84**	**0.84**
Radiology-llama2 7B	0.77	0.82	0.77	0.76
llama-3.1 8B	0.70	0.73	0.70	0.65

Results

- **Fine-tuning**
  : 의료자문 1차 - Label 재정의
  (Medllama3-v20 8B
  - hyperparameter tuning)



Settings	accuracy	precision	recall	f1 score
len1024_4bit_early stop	0.84	0.88	0.83	0.79
len1024_4bit_full epoch	**0.92**	**0.92**	**0.92**	**0.92**
len1024_8bit_full epoch	Out of Memory in server			
len600_4bit_full epoch	0.88	0.89	0.88	0.88
len600_8bit_full epoch	0.90	0.91	0.90	0.89

- # Prompt engineering
  : <u>의료자문 2차</u> – **Report 분석 순서 명기**

Prompt engineering	**accuracy**	**precision**	**recall**	**f1 score**
1. Diagnosis process	0.93	0.94	0.93	0.93
2. Diagnosis process + CoT	0.94	0.95	0.94	0.94
3. Diagnosis process + Note	0.91	0.93	0.91	0.91
4. Diagnosis process + Note + CoT	0.91	0.93	0.91	0.91
5. Diagnosis process + CoT + rule-out	**0.95**	**0.95**	**0.95**	**0.95**
6. Diagnosis process + CoT + radiologist	0.95	0.95	0.95	0.95
7. Diagnosis process + CoT + rule-out + example & keyword	**0.90**	**0.91**	**0.89**	**0.89**

- Prompt engineering
  : **Modified – 전체 report 보기**

Prompt engineering	**accuracy**	**precision**	**recall**	**f1 score**
1. Diagnosis process	0.93	0.94	0.93	0.93
2. Diagnosis process + CoT	0.93	0.94	0.93	0.93
3. Diagnosis process + Note	0.91	0.93	0.91	0.91
4. Diagnosis process + Note + CoT	0.91	0.93	0.91	0.91
5. Diagnosis process + CoT + rule-out	**0.93**	**0.94**	**0.93**	**0.93**
6. Diagnosis process + CoT + radiologist	0.94	0.95	0.94	0.94
7. Diagnosis process + CoT + rule-out + example & keyword	**0.88**	**0.90**	**0.87**	**0.87**

Results

- Base 5 (Best) vs. Base 7

Baseline prompt + Diagnosis process + CoT + Rule-Out(R/O)

Instruction
Based on the report below,.....

- Description of six labels
  …
  - romets
    -**R/O** should be interpreted as romets.

Focus on the key terms in the "Conclusion" and "Impression" section

1. Focus on the "Conclusion" section for primary classification.
2. If needed, refer to the "Findings" section…
3. Ignore sections that do not contribute to the classification, such as patient history.
4. Pay special attention to key terms … such as **"R/O"**, …
5. Provide final the classification output only.

Let's think step by step.

Baseline prompt + Diagnosis process + Example & Keyword

Instruction
Based on the report below,.....

- Description of six labels
  - no
    - This label indicates the absence of any detectable abnormality or condition in the imaging study.
    - **Example:** "No evidence of metastasis or tumor recurrence." or "No abnormal findings."...
    - **Keywords:** "no evidence", "no evidence of metastasis", "no abnormal"

…

  - romets
    - **Example:** "Follow-up imaging recommended to rule out metastasis."...
    - **Keywords**: "recommendation: clinical correlation", "follow up", "r/o", "possibility"

Focus on the key terms in the "Conclusion" and "Impression" section

…

- Base 5 (Best) vs. modified 5

Baseline prompt +
Diagnosis process + CoT + Rule-Out(R/O)

Instruction
Based on the report below,.....

- Description of six labels
  …
  - romets
    -R/O should be interpreted as romets.

Focus on the key terms in the "Conclusion" and "Impression" section
1. Focus on the "Conclusion" section for primary classification.
2. **If needed**, refer to the "Findings" section…
3. Ignore sections that do not contribute to the classification, such as patient history.
4. Pay special attention to key terms …
5. Provide final the classification output only.

Let's think step by step.

Baseline prompt +
Diagnosis process + CoT + Rule-Out(R/O)

Instruction
Based on the report below,.....

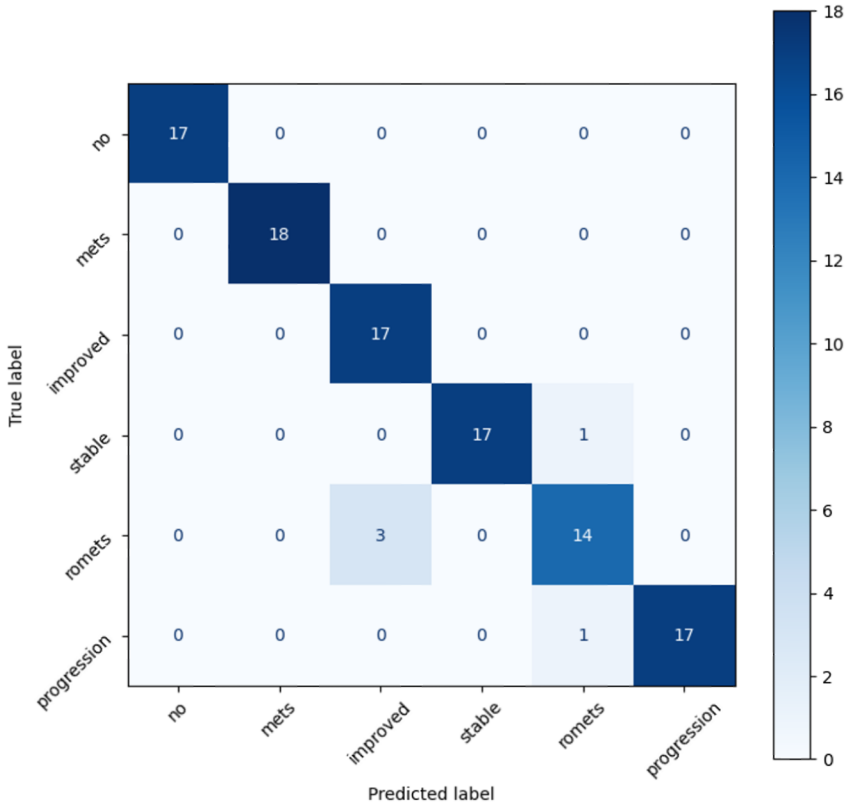
- Description of six labels
  - no
  - mets
  ….
  - romets
    R/O should be interpreted as romets.

Focus on the key terms in the "Conclusion" and "Impression" section
1. Focus on the "Conclusion" section for primary classification.
2. **Second**, refer to the "Findings" section for additional context.
3. Ignore sections that do not contribute to the classification, such as patient history.
4. Pay special attention to key terms indicating the patient's condition, such as "stable," "metastasis," "progression," "no evidenfinal," "improved," and similar phrases.
5. Provide final_the classification output only.

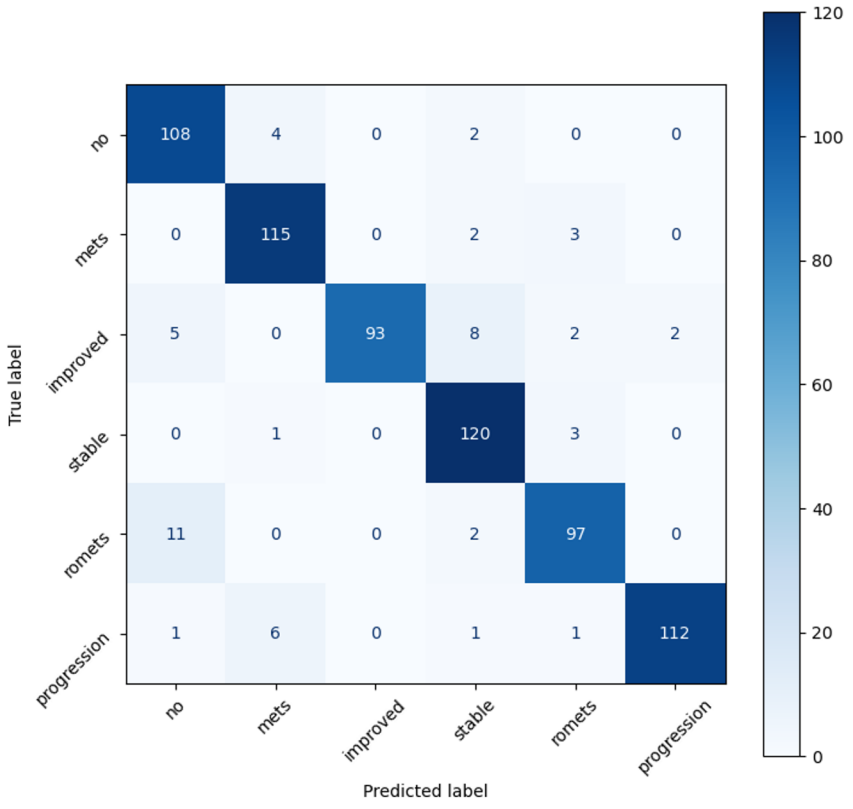
Let's think step by step.

Results

- Cross Validation

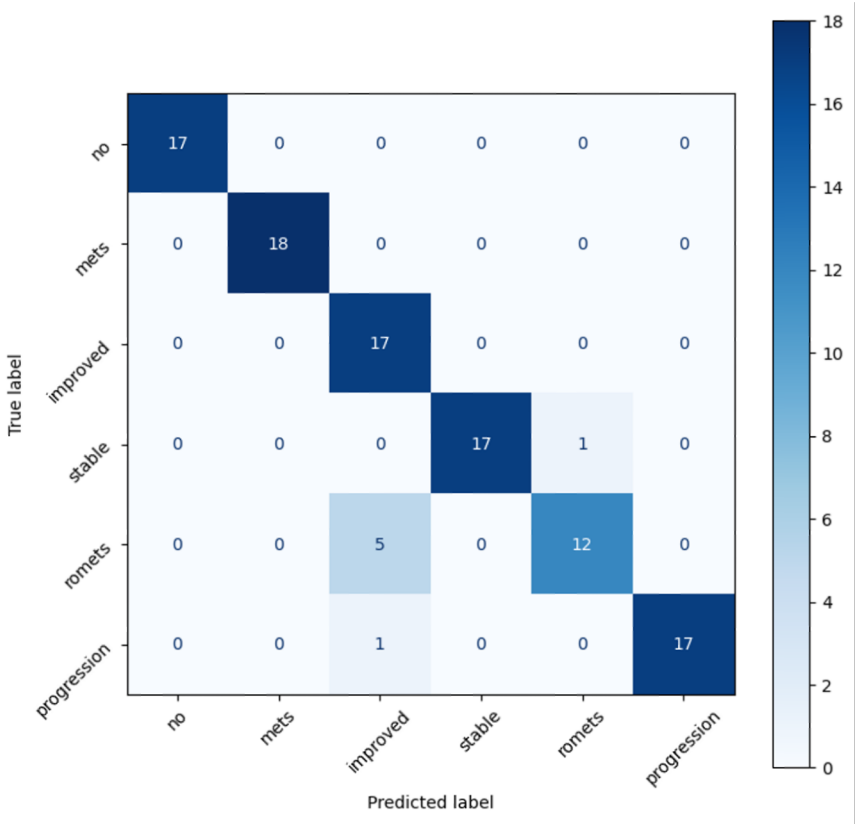


confusion matrix for
“5. Diagnosis process + CoT + rule-out” prompt

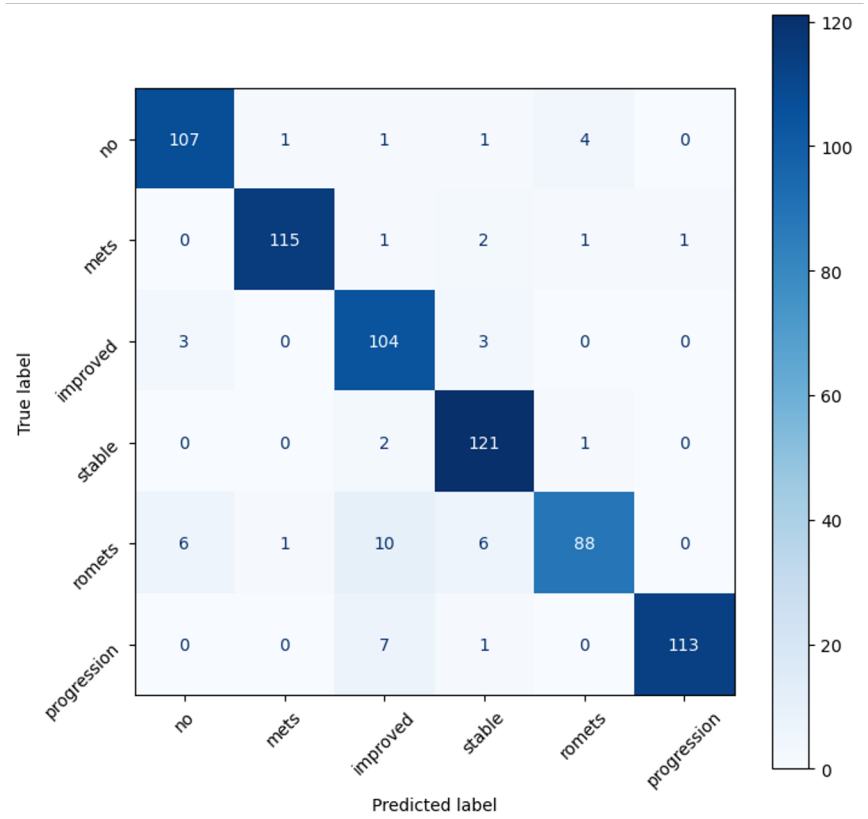


confusion matrix for
“5. Diagnosis process + CoT + rule-out” prompt
with **5-fold** cross validation

- Cross Validation – modified prompt



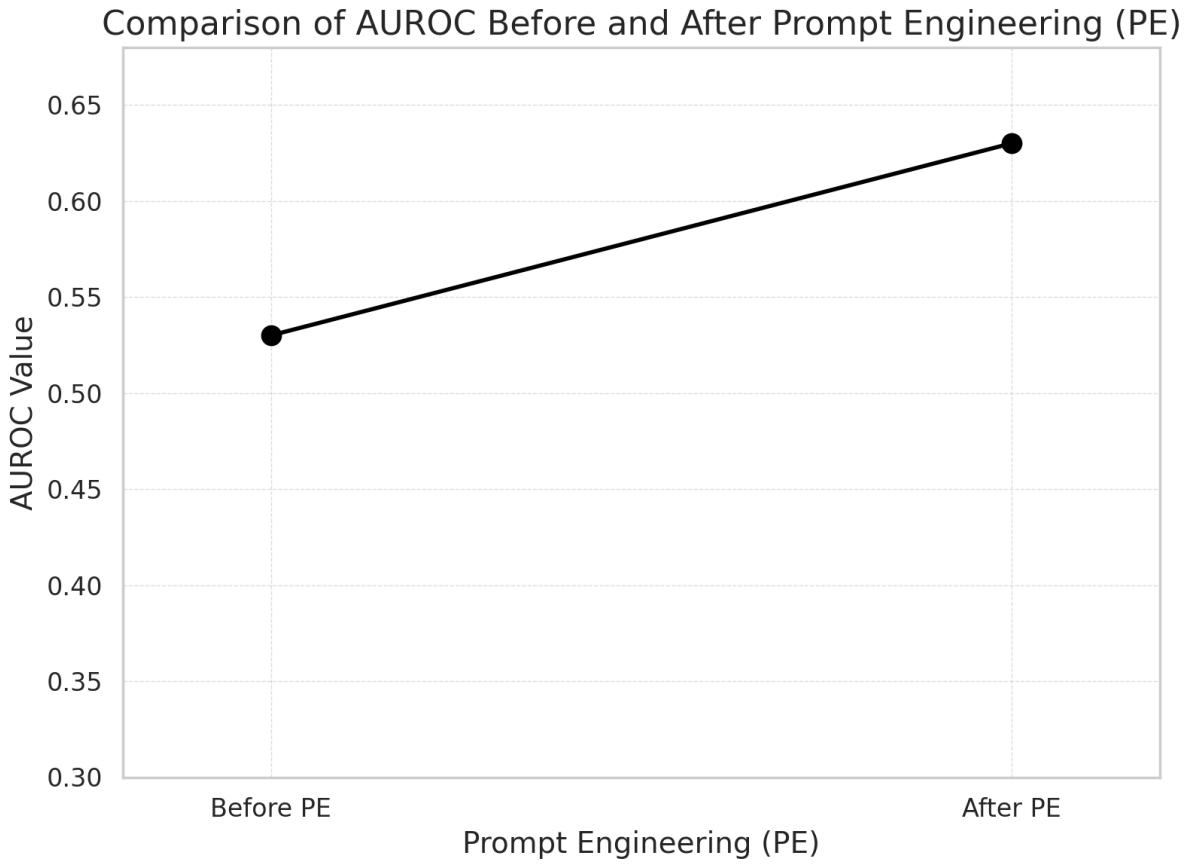
confusion matrix for
“5. Diagnosis process + CoT + rule-out” prompt



confusion matrix for
“5. Diagnosis process + CoT + rule-out” prompt
with **5-fold** cross validation

Results

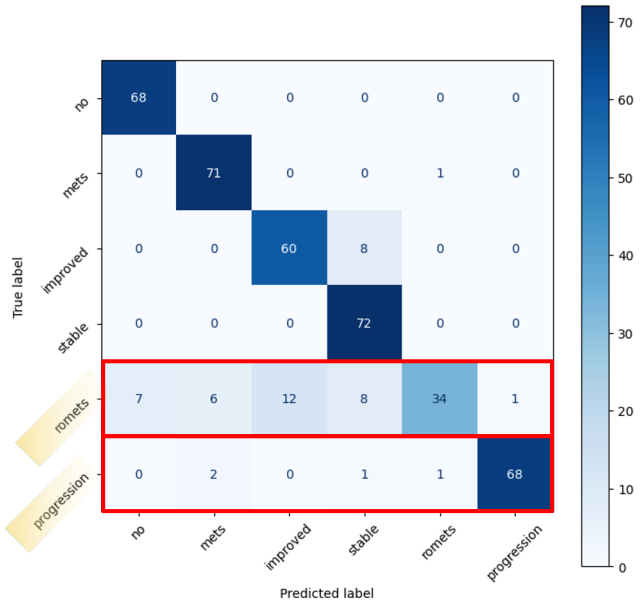
- Confidence score based on AUROC



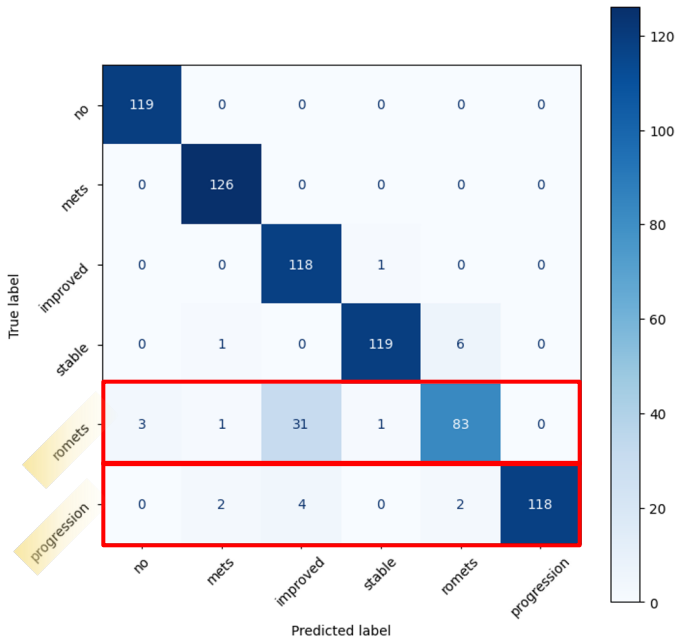
* PE: Prompt engineering

Input label and report

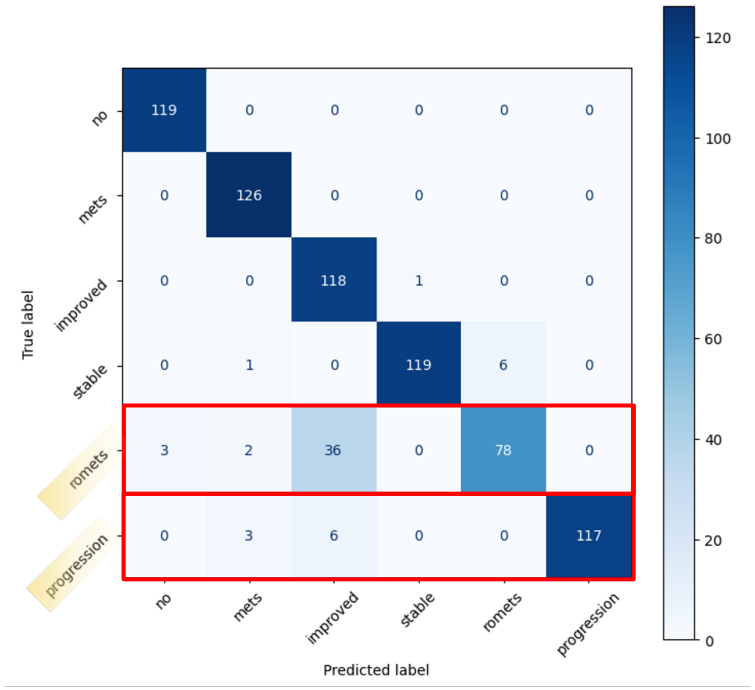
- Hallucination analysis - Ensemble approach



sum of best 4 hyperparameter tuned models



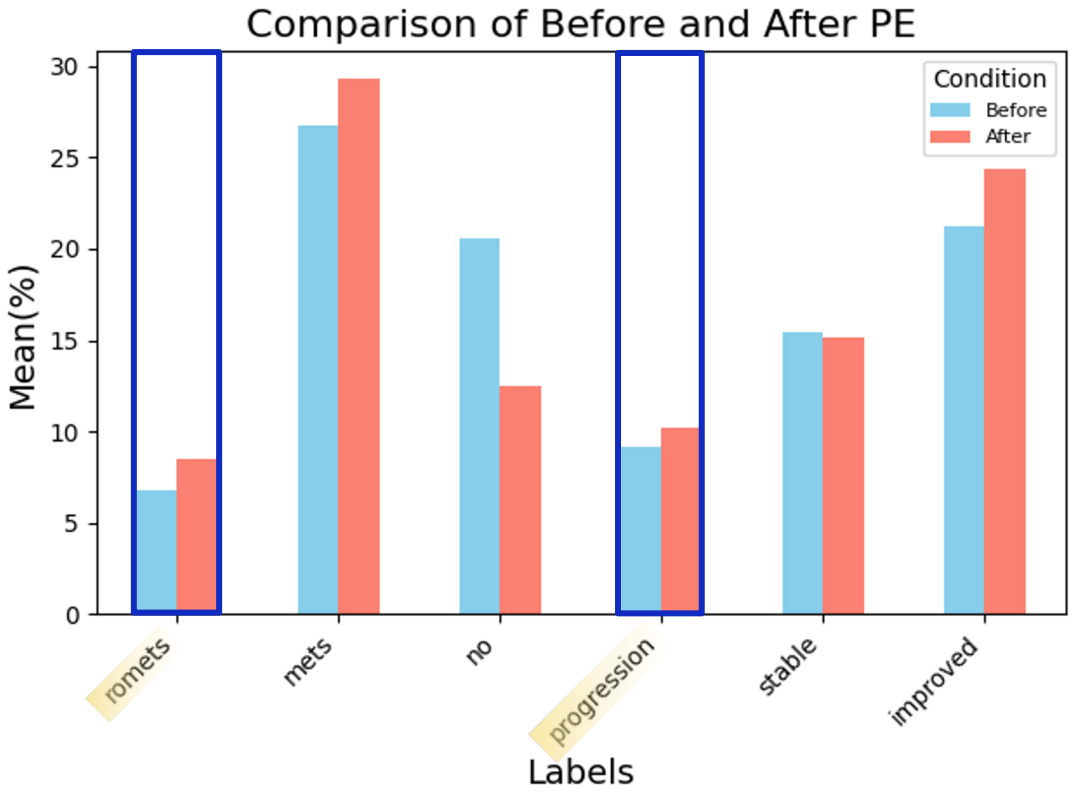
sum of best 7 prompts



sum of modified-best 7 prompts

Results

- Hallucination analysis - average confidence scoring approach



Input label and report

Hallucination analysis - qualitative approach

- Romets를 improved 라고 잘못 예측한 경우
  - romets : 전이가 의심되어 추가 조사가 필요함
  - improved : 환자의 상태나 증상이 좋아졌음을 의미

Report (romets)

EXAM: Whole spine MRI with contrast enhancement and metal artifact reduction.
Clinical information: Upper limb weakness, s/p decompressive laminectomy and posterior fixation, s/p corpectomy and anterior interbody fixation due pathologic fracture T2 (Osteosarcoma), s/p Radiotherapy.

Findings:
s/p Decompressive laminectomy and posterior fixation, T1-2-3-T4-5.
s/p Anterior thoracic corpectomy and fusion T2-3.
Swelling and T2 hyperintensity with enhancement in posterior back muscle and subcutaneous fat layer of thoracic level.
Epidural extension and encroachment of both foraminal canal at T2/3.
Bone marrow signal change and enhancement in T2 and T3.
Diffuse marrow signal changes including thoracic vertebrae.
Clinical significance indeterminate.

Conclusions:
s/p Decompressive laminectomy and posterior fixation, T1-2-3-T4-5.
s/p Anterior thoracic corpectomy and fusion T2-3.
Decreased epidural extent compared with 2021-12-16 MRI.
R/O Remained lesion, T2 and T3.

Recommendation:
Clinical correlation and follow up.

LLM response (improved)

```
Explanation:

Improved epidural extent compared with 2021-12-16 MRI.
R/O Remained lesion, T2 and T3.
```

Hallucination analysis - qualitative approach

- Stable을 romets 라고 잘못 예측한 경우
  - romets : 전이가 의심되어 추가 조사가 필요함
  - stable : 이전 영상 검사 이후 상태나 소견에 큰 변화가 없음 의미

Report (stable)

Exam: Whole spine MRI with contrast enhancement.
Clinical information: Breast cancer, Rt. Local recur,

Compared with outside MRI taken on 2021.10.06

Findings:
No remarkable change of focal one marrow replacing and enhancing lesion in T9 vertebral body,
R/O bone metastasis.
No definite spinal cord signal change or enhancing lesion.
No evidence of pathologic fracture in this study.
No newly developed lesion.
OPLL at T4-5-6.
Disc bulging at L4/5.

Impression:
R/O Bone metastasis in T9 body without interval change.

Recommendation:
Clinical correlation.

LLM response (romets)

```
Explanation:

R/O Bone metastasis in T9 body without interval change.
No remarkable change of focal one marrow replacing lesion in T9 vertebral body.
No evidence of pathologic fracture in this study.
No newly developed lesion.
```

- Hallucination analysis - qualitative approach
  - Progression을 romets 라고 잘못 예측한 경우
    - romets : 전이가 의심되어 추가 조사가 필요함
    - progression : 질병이 악화되거나 진행되고 있다는 증거가 있음을 의미

Report (progression)

EXAM : Whole spine MRI with contrast enhancement
HISTORY : Prostate cancer with multiple bone metastases

Compared to previous MRI taken on 2019.08.16,

FINDING :
Newly developed bone marrow replacing lesions with enhancement in T1 vertebral body, left lamina and pedicle of L5, left sacrum and ilium.
Not well delineation of previous noted enhancing lesion in the lamina of C2.
Spondylolisthesis of C4 over C5.
Retrolisthesis of C6 over C7.
Mild kyphotic deformity and mild central canal stenosis on C3-4-5-6-7 level.

Disc bulging in T12/L1, L2/3, L3/4, and L4/5.
- Mild central canal narrowing and bilateral neural foraminal narrowing in L3/4.
- Bilateral moderate neural foraminal narrowing in L4/5.

Spondylolytic spondylolisthesis of L5 over S1.
- Bilateral severe neural foraminal narrowing.

IMPRESSION :
1. **Probable bone metastases in T1 vertebral body, left lamina and pedicle of L5, left sacrum and left ilium.**
2. Not well delineation of previous noted enhancing lesion in the lamina of C2.
3. Spondylolytic spondylolisthesis of L5 over S1 with bilateral severe neural foraminal narrowing.

RECOMMEND : clinical correlation.

LLM response (romets)

```
Explanation:

R/O Bone metastasis in T1 vertebral body, left lamina and pedicle of L5, left sacrum and ilium.
- Not well delineation of previous noted enhancing lesion in the lamina of C2.
- Slightly increased extent of bone marrow replacing lesions in T1 vertebral body, left lamina an
- Pathologic compression fracture in T1 vertebral body.
- No evidence of central canal stenosis or neural foraminal stenosis.
```

Conclusion

Conclusion

- Summary

  - 본 연구에서는 암 치료 반응 분류를 위한 LLM 기반 모델을 구축하고, **다양한 프롬프트 엔지니어링과 모델 파라미터 튜닝**을 통해 성능을 최적화했음.

  - Medllama3-v20 8B 모델을 선정하여 사전 학습된 LLM을 사용해 **다양한 환경에서 fine-tuning을 진행**한 결과, max sequence length 1024 + 4bit + full epoch 설정이 accuracy, precision, recall, f1 score 모두에서 **0.92**로 가장 우수한 성능을 보였음.

  - 또한, 여러 프롬프트 디자인을 비교한 결과 "**Diagnosis process + CoT + rule-out**" 프롬프트가 모든 성능 지표에서 **0.95**로 가장 높은 성능을 보였음.

Conclusion

- Strength
  - 특히, **임상의 자문**을 반영한 프롬프트 구성과 세밀한 파라미터 튜닝이 결과의 **정확도와 신뢰도**를 높이는 데 기여했다고 할 수 있음.
  - LLM 환경에서의 AUROC 점수를 확보하기 위해 **confidence score based on AUROC** 방법론을 새롭게 도입함.
  - Ensemble, average confidence scoring approach를 활용하여 LLM 환경에서의 **explainability 확보 및 hallucination 완화**를 위한 지표를 구성함.
  - **오답의 원인을 분석**하기 위해 LLM에게 오답인 경우에 한해 **추론 과정을 되묻는 프롬프트**를 설계함.

Conclusion

- Future work

  - 프롬프트 디자인의 길이가 길어지는 경우 성능 저하를 초래함.

  - 8B와 같은 작은 llm 모델은 한 번에 단순한 내용의 지시만 수행 가능함.
    추후 70B와 같은 거대 파라미터 모델을 활용하여 성능을 높이는 연구가 필요함.

  - 특정 라벨에 대한 낮은 예측 성능을 완화하기 위해 더 많은 데이터를 확보하여
    추가 연구가 필요함.

Q & A

Thank you

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- Hallucination – confidence score
  - Average confidence scoring approach of the input label and report

	Input **label and report**		Input only **report**	
	Before PE(%)	After PE(%)	Before PE(%)	After PE(%)
romets	6.83	8.51	12.37	7.87
mets	26.75	29.32	20.99	24.80
no	20.59	12.47	31.69	36.43
progression	9.13	10.18	6.60	8.79
stable	15.42	15.13	18.69	13.60
improved	21.25	24.35	9.63	8.46

average confidence scoring approach